DESIGN NOTE:
Potential design error: Unless it's present on the main board, there should be a snubber diode in parallel with the speaker coil.

DESIGN NOTE:
Buzzer is an equivalent substitute based physical and electrical measurements.

DESIGN NOTE:
All ferrite beads are substitutes based on VNA scans.
See DOC/Ferrite Beads VNA.ods for source data.

J3
FF14-12A-R11BL-3J
MH
TP1
1 BUZZER
2 PP_3V3_CONN
3 D_GND_CONN
4 HP_RIGHT_CONN
5 HP_LEFT_CONN
6 AUDIO_GND_CONN
7 HP_DETECT
8 UART_REMOTE_RX_CONN
9 UART_REMOTE_TX_CONN
10 HOLD_SW
11
12
MH
GND

TP2
R11 8,2 1% 63mW
BUZZER_R
LS1 CSS-0578-SMT-TR
TP3
TP4
TP5
S1 SK-12C0405-SG 1.5 RT
SH
GND
TP8

TP6
C7 100uF 4V TANT 10%
HP_RIGHT_CAP
R151 3,3 1% 63mW
HP_RIGHT_R
TP11
TP7
C5 100uF 4V TANT 10%
HP_LEFT_CAP
R145 3,3 1% 63mW
HP_LEFT_R
TP12
R143 3.9k 1% 63mW
R144 3.9k 1% 63mW
TP9
TP10

L8 BLM15AG121SN1D 120 Ohm @ 100 MHz 550 mA 190mOhms HP_RIGHT_FILT
L9 BLM15AG121SN1D 120 Ohm @ 100 MHz 550 mA 190mOhms HP_LEFT_FILT
L10 MU1608-151Y 150 Ohms @ 100 MHz 600 mA 250mOhms AUDIO_GND_FILT
L4 MMZ1005Y102CT000 1kOhm @ 100 MHz 200 mA 700mOhms PP_3V3_FILT
L5 MMZ1005Y102CT000 1kOhm @ 100 MHz 200 mA 700mOhms D_GND_FILT
L6 MMZ1005Y102CT000 1kOhm @ 100 MHz 200 mA 700mOhms UART_REMOTE_RX_FILT
L7 MMZ1005Y102CT000 1kOhm @ 100 MHz 200 mA 700mOhms UART_REMOTE_TX_FILT

C11 100pF 50V NP0/C0G 5%
C12 100pF 50V NP0/C0G 5%
C13 100pF 50V NP0/C0G 5%
C14 100pF 50V NP0/C0G 5%
C8 100pF 50V NP0/C0G 5%
C9 100pF 50V NP0/C0G 5%
C10 100pF 50V NP0/C0G 5%

L15 BLM15AG121SN1D 120 Ohm @ 100 MHz 550 mA 190mOhms
L16 BLM15AG121SN1D 120 Ohm @ 100 MHz 550 mA 190mOhms
L17 0 Jumper
L11 MMZ1005Y102CT000 1kOhm @ 100 MHz 200 mA 700mOhms
L12 MMZ1005Y102CT000 1kOhm @ 100 MHz 200 mA 700mOhms
L13 MMZ1005Y102CT000 1kOhm @ 100 MHz 200 mA 700mOhms
L14 MMZ1005Y102CT000 1kOhm @ 100 MHz 200 mA 700mOhms

HP_TVS
DZP1 BZX84C6V2TS-7-F 6.2V -
DZP2 BZX84C6V2TS-7-F 6.2V -
TP13

J1
IPOD 4TH GEN COMBO JACK
HP_RIGHT R1
HP_DETECT SW
HP_LEFT T
R2
AUDIO_GND S
PP_3V3_REMOTE 1
D_GND 2
UART_REMOTE_RX 3
UART_REMOTE_TX 4
SH
GND

DESIGN NOTE:
Spring contacts are an equivalent substitute based physical measurements.
SP1 S7041-42R
SP2 S7041-42R
GND

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Title: iPod 4th Gen. B/W HP/Hold Board			Part Number: 820-1635
Root			
Size: A3	Date: 2026-01-25	DWG Rev: A	**Rev: A**
File: 820-1635.kicad_sch			Sheet: 1/1